# Kitchen Privacy Screen — Proposal

Cornelius, NC · November 10, 2026 · valid for 30 days · two options

## Scope (both options)

- 22'-1" long, 8'-0" tall horizontal privacy screen with pergola top, finished on both sides, matching the reference image.
- **Posts:** pressure-treated pine 4x4 core (ground contact), set **24" deep** in concrete over 6" of gravel, holes dug with a rented mini skid steer and auger. Above grade, each post is clad in cedar: 2x6 faces and 1" side boards, all 5 1/2" wide with flush faces. The buried part is treated pine only, for lower cost and longer life in the soil.
- **Slats:** cedar 2x4 laid flat, held in notched side boards (even spacing, no visible fasteners on the slat faces).
- **Pergola top:** notched cedar 2x8 top boards on both sides, end caps for a solid-beam look, cedar 2x6 rafters with cloud-lift ends, one per post.
- **Fasteners:** coated trim-head deck screws (small countersunk heads, rated for treated lumber) and 1/2" galvanized through-bolts.
- All visible parts are pre-cut, notched and eased in our shop to keep on-site work short. Cleanup and debris removal included.

## Comparison

| | Option A — 10 posts | Option B — 7 posts |
|---|---|---|
| Posts | 10 | 7 |
| Post spacing (on center) | 2'-4 13/16" | 3'-7 1/4" |
| Clear bay (slat span) | 23 5/16" | 3'-1 3/4" |
| Warping risk | Low (< 36") | Moderate (> 36") |
| Slats | 144 | 96 |
| Materials | $4,121 | $3,436 |
| Labor (shop + site) | $3,940 (66 h) | $3,045 (51 h) |
| Service fee | $150 | $150 |
| **Total** | **$8,211** | **$6,631** |

Option A uses the golden-ratio spacing with the shortest slat span (best against warping). Option B uses fewer posts and fewer parts (lower cost), with longer slats.

## Materials and finish

| Part | Material (species, grade, treatment) | Surface | Actual size | Qty A 10 posts | Qty B 7 posts |
|---|---|---|---|---|---|
| Post core (buried 24") | Pressure-treated Southern Yellow Pine 4x4, #2, ground contact (UC4B) | S4S (planed) | 3 1/2" x 3 1/2" x 10' | 10 | 7 |
| Post faces (house + neighbor) | Western Red Cedar 2x6, kiln-dried | S4S (planed) | 1 1/2" x 5 1/2" x 7'-5 1/2" | 20 | 14 |
| Notched side boards | Western Red Cedar 5/4x6, kiln-dried, ripped to 3 1/2" | S4S, eased edges | 1" x 3 1/2" x 7'-5 1/2" | 18 | 12 |
| Plain side boards (end posts) | Western Red Cedar 5/4x6, kiln-dried, ripped to 3 1/2" | S4S, eased edges | 1" x 3 1/2" x 7'-5 1/2" | 2 | 2 |
| Slats (laid flat) | Western Red Cedar 2x4, kiln-dried | S4S (planed) | 1 1/2" x 3 1/2" x 2'-1 1/16" (A) / 3'-3 1/2" (B) | 144 | 96 |
| Top boards (notched for rafters) | Western Red Cedar 2x8, kiln-dried | S4S (planed) | 1 1/2" x 7 1/4" (3 pieces per side, 8') | 6 | 6 |
| Top-board end caps | Western Red Cedar 2x8, kiln-dried | S4S (planed) | 1 1/2" x 7 1/4" x 9 1/2" | 2 | 2 |
| Pergola rafters (cloud-lift ends) | Western Red Cedar 2x6, kiln-dried | S4S (planed) | 1 1/2" x 5 1/2" x 36" | 10 | 7 |

**Finish:** all cedar is Western Red Cedar, kiln-dried and **S4S (surfaced/planed on 4 sides)**, so every piece has true, consistent dimensions and the post faces, side boards and notches line up. Rough-sawn cedar (as stocked at some big-box stores in 2x6/2x8) is not used; if S4S is not stocked locally, it will be sourced from a lumber yard. Cedar is left natural (no stain needed) and will weather to silver-gray; sealer available on request.

## Option A — 10 posts

- **10 composite posts** at 2'-4 13/16" on center, clear bay 23 5/16" (bay-to-post ratio 4.24 ≈ $\varphi^3$ (Fibonacci 55/13)). Slat span is well under the 36" limit we use to prevent warping.
- **144 cedar slats** (16 per bay, 1 11/16" gaps), 2'-1 1/16" long, held in notched side boards.
- **Pergola top:** 6 notched cedar top boards, 2 end caps, 10 rafters with cloud-lift ends.
- Drawings: sheets A-1 to A-3 at the end of this document.

| | |
|---|---|
| Materials (incl. 8.25% Mecklenburg sales tax and equipment rental $450) | $4,121 |
| Labor — shop pre-cutting 38 h + on-site installation 28 h (66 h × $60/h) | $3,940 |
| Service fee | $150 |
| **Total Option A** | **$8,211** |

## Option B — 7 posts

- **7 composite posts** at 3'-7 1/4" on center, clear bay 3'-1 3/4" (bay-to-post ratio 6.86 ≈ $\varphi^4$ (Fibonacci 89/13)). Slat span (3'-1 3/4") is above our conservative 36" limit (still under the 48" maximum for 2x4 cedar): slightly more risk of bowing over time than Option A.
- **96 cedar slats** (16 per bay, 1 11/16" gaps), 3'-3 1/2" long, held in notched side boards.
- **Pergola top:** 6 notched cedar top boards, 2 end caps, 7 rafters with cloud-lift ends.
- Drawings: sheets B-1 to B-3 at the end of this document.

| | |
|---|---|
| Materials (incl. 8.25% Mecklenburg sales tax and equipment rental $450) | $3,436 |
| Labor — shop pre-cutting 27 h + on-site installation 23 h (51 h × $60/h) | $3,045 |
| Service fee | $150 |
| **Total Option B** | **$6,631** |

## Optional extra (not included)

**Landscaping strip** along the neighbor side (22 ft): landscape fabric, edging and river rock border: **$351** (materials + labor), same for both options.

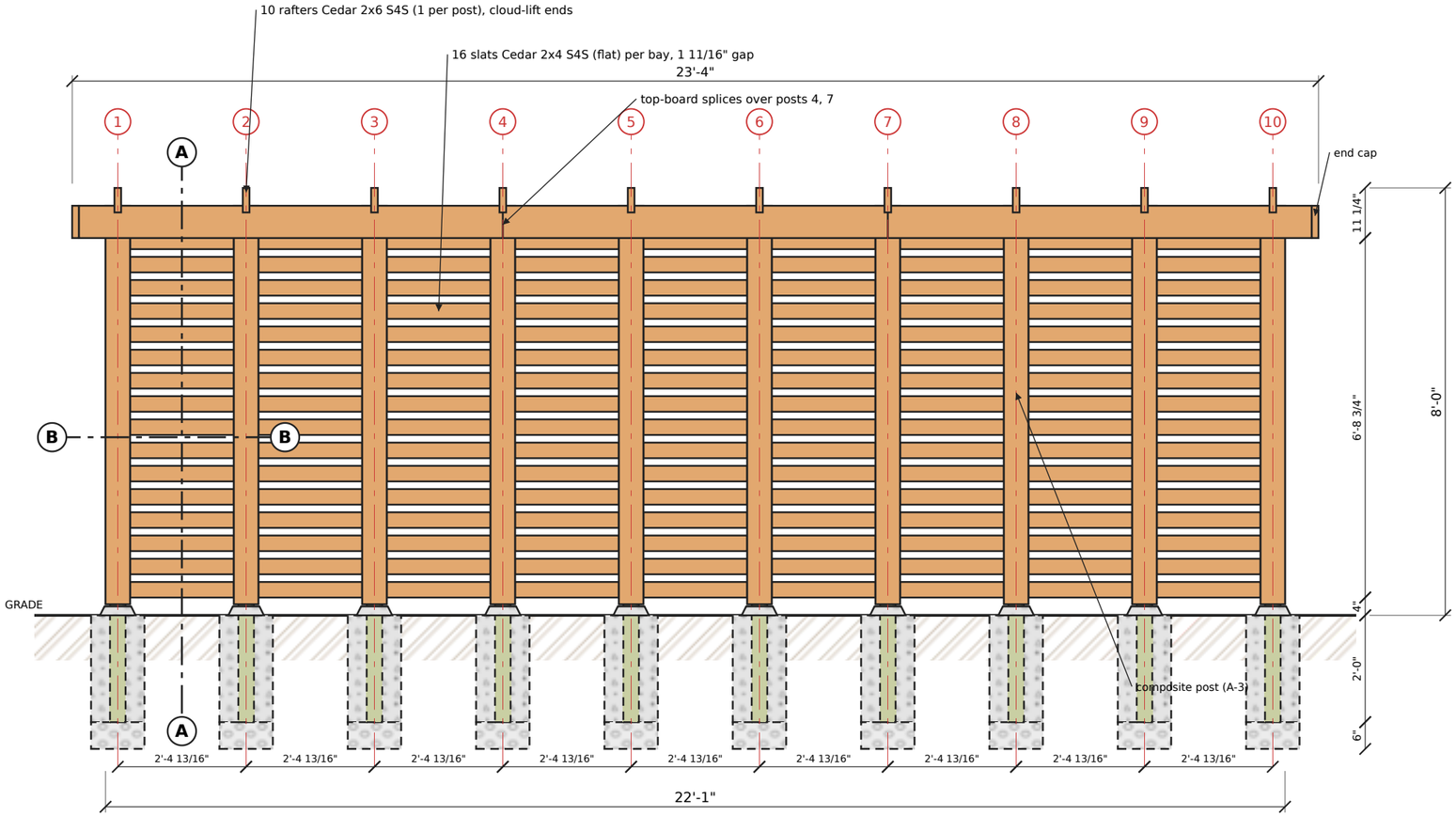
## Installation sequence (both options)

1. Holes (machine auger)
2. Posts (treated pine cores in concrete)
3. Level post tops
4. Notched side boards (pre-cut), from the neighbor side
5. Slats (pre-cut)
6. Front closure (2x6 faces)
7. Side closure (end posts)
8. Top boards (pre-notched)
9. End caps (pre-cut)
10. Pergola rafters (pre-cut)

## Notes, exclusions and payment

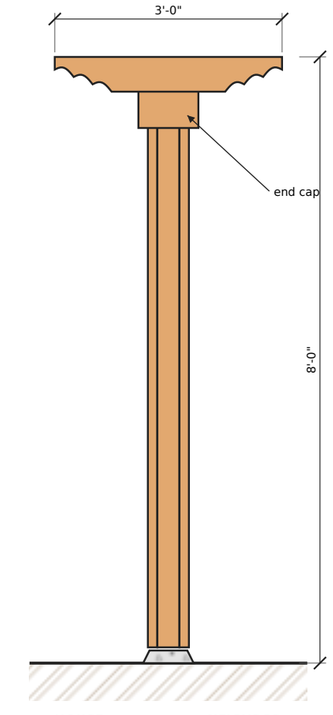
- Material prices are based on current Lowe's / Home Depot pricing and lumber-yard estimates; they will be confirmed at purchase. Changes above 5% may be adjusted.
- Not included: landscaping strip (optional extra above), permits/HOA fees, surveys, rock or root removal beyond normal digging, plants/vines. We will call NC811 before digging.
- Payment: 40% deposit to order materials and start shop work · 40% when posts and slats are installed · 20% on completion.

Drawings follow: Option A (10 posts) sheets A-1 to A-3, then Option B (7 posts) sheets B-1 to B-3.



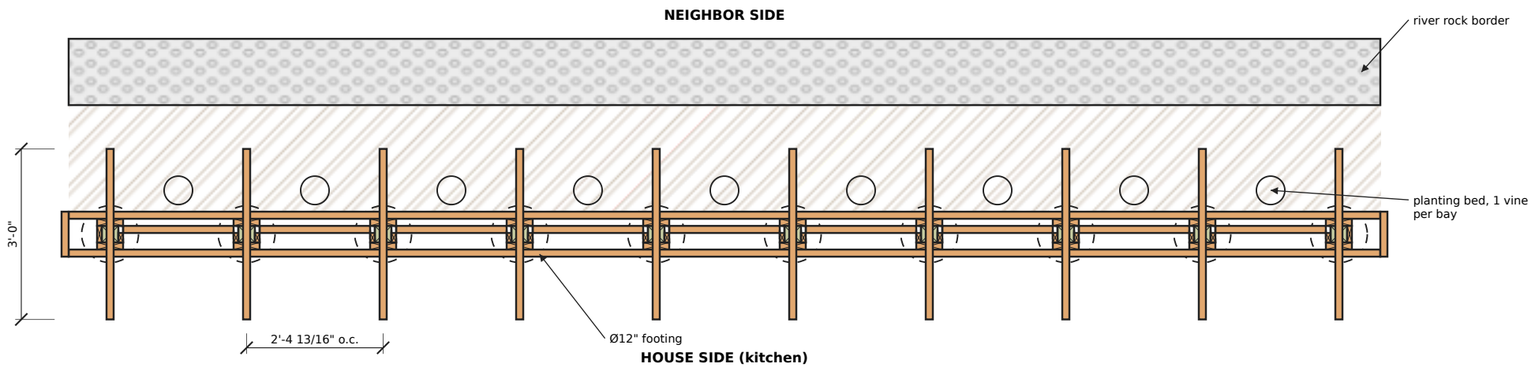

## ELEVATION — NEIGHBOR SIDE

SCALE 1:32 (3/8" = 1'-0")




## SIDE ELEVATION (end view)

SCALE 1:24 (1/2" = 1'-0")




## PLAN — TOP VIEW

SCALE 1:32 (3/8" = 1'-0")

**NOTES**

1. Overall length 22'-1"; 10 posts at 2'-4 13/16" o.c.; clear bay 23 5/16".
2. Bay/post = 4.24 ≈ φ^3 (Fibonacci 55/13); span 23 5/16" < 36" (low warping risk).
3. Post cores embedded 2'-0" in concrete over 6" gravel; holes by machine auger.
4. Call NC811 before digging. Verify property line, setbacks and HOA.
5. Fasteners: coated trim-head deck screws (ACQ-rated), countersunk.
6. 16 slats per bay, 1 11/16" gap set by the notched side boards.
7. All posts 5 1/2" wide, faces flush. Parts marked YES are pre-cut in the shop.
8. Cedar = Western Red Cedar, kiln-dried, S4S (planed 4 sides).
9. PT = Southern Yellow Pine #2, ground contact.

**Kitchen Privacy Screen — 10 posts, cedar with pergola top (Cornelius, NC)**

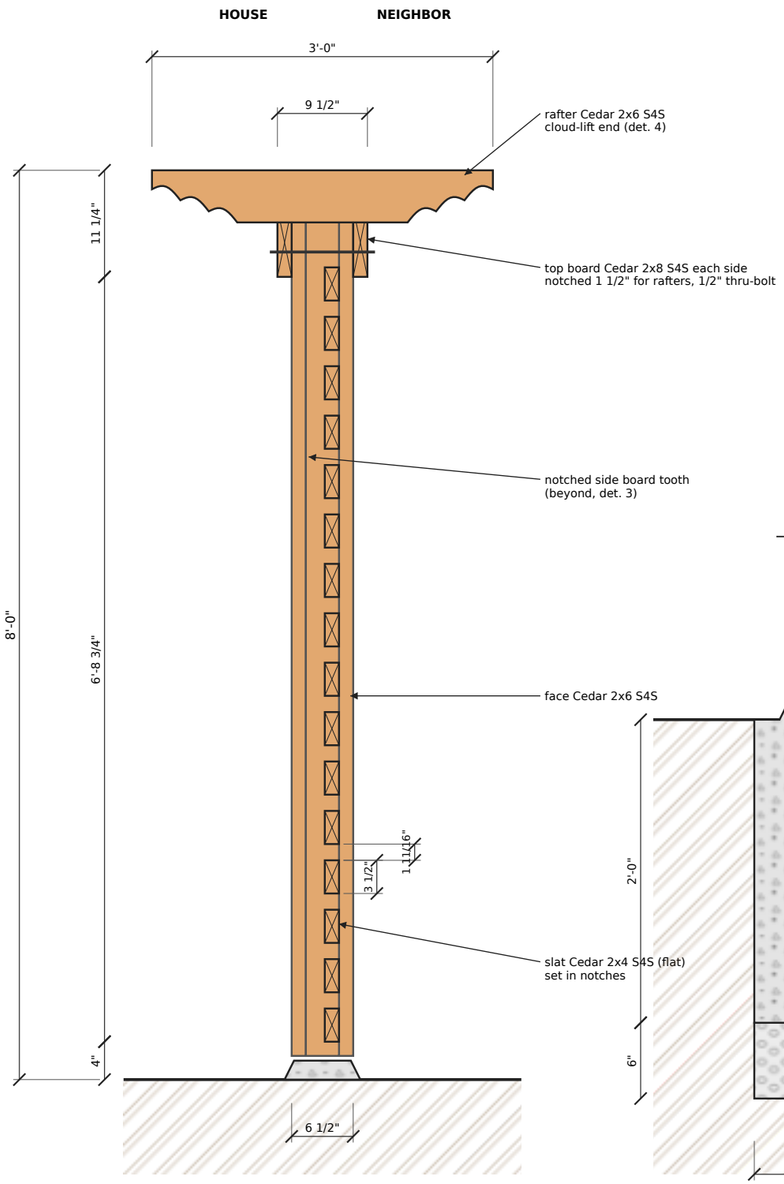
**Elevation and plan**

Date: 2026-11-10 Rev.: 7 By:

Dimensions in feet/inches. Lumber: actual sizes (2x6 = 1 1/2" x 5 1/2").

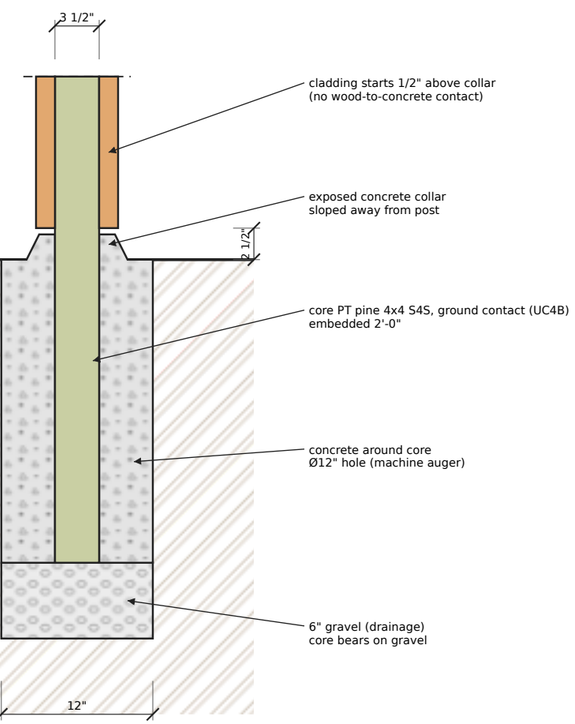
Verify all dimensions in the field before cutting. Not a structural design.

**A-1**



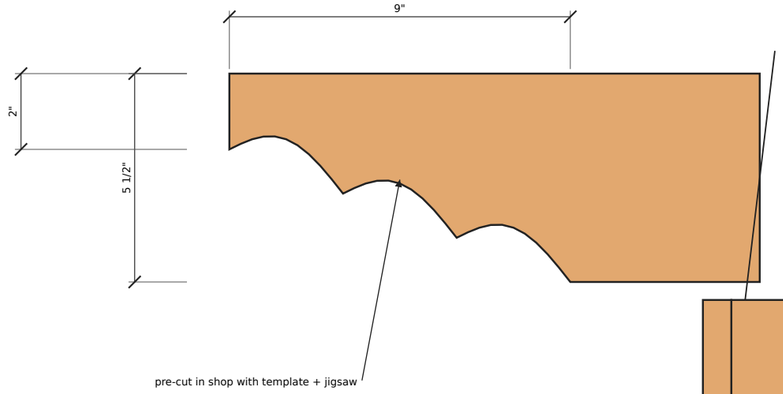

## SECTION A-A (mid-bay, looking at post 1)

SCALE 1:16 (3/4" = 1'-0")



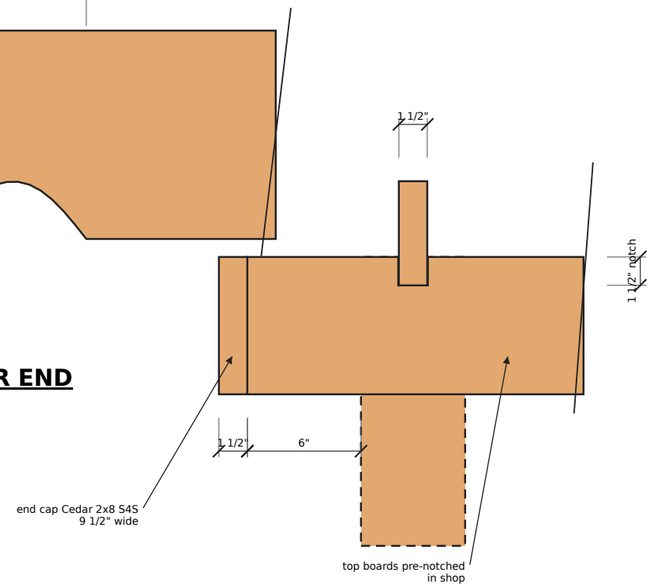

## DETAIL 1 — POST FOOTING

SCALE 1:12 (1" = 1'-0")



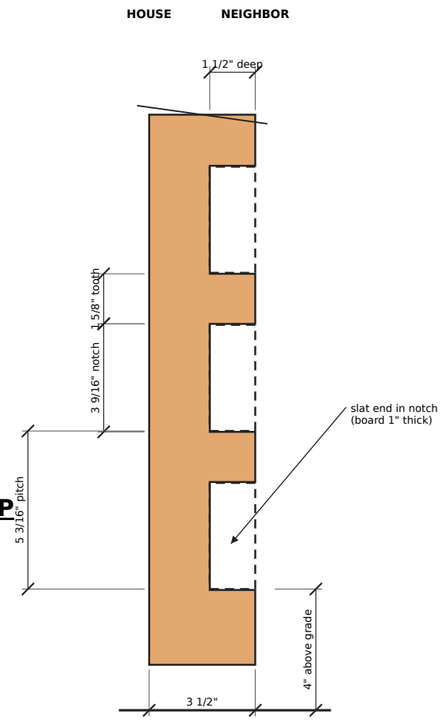

## DETAIL 4 — RAFTER END

SCALE 1:4 (3" = 1'-0")




## DETAIL 5 — TOP BOARD END: NOTCH + END CAP

SCALE 1:8 (1 1/2" = 1'-0")




## DETAIL 3 — NOTCHED SIDE BOARD (from bay)

SCALE 1:5

**PARTS LIST**

| Mark | Part | Material | Qty | Length | Shop pre-cut / notes |
|---|---|---|---|---|---|
| N | Post core | PT pine 4x4 S4S, ground contact (UC4B) | 10 | 9'-8" | embed 2'-0"; trim tops on site |
| S | Notched side board | Cedar 5/4 S4S (1") notched | 18 | 7'-5 1/2" | YES: rip to 3 1/2", 16 notches |
| R | Slat | Cedar 2x4 S4S (flat) | 144 | 2'-1 1/16" | YES: cut to length (stop block) |
| F | Post face | Cedar 2x6 S4S | 20 | 7'-5 1/2" | YES: cut to length |
| L | Plain side (end posts) | Cedar 5/4 S4S (1") plain | 2 | 7'-5 1/2" | YES: rip to 3 1/2" |
| T | Top board (3 per side) | Cedar 2x8 S4S | 6 | 7'-11 1/4", 7'-2 1/2" | YES: notches for rafters |
| E | Top board end cap | Cedar 2x8 S4S | 2 | 9 1/2" | YES |
| K | Pergola rafter | Cedar 2x6 S4S | 10 | 3'-0" | YES: cloud-lift ends |

**Kitchen Privacy Screen — 10 posts, cedar with pergola top (Cornelius, NC)**

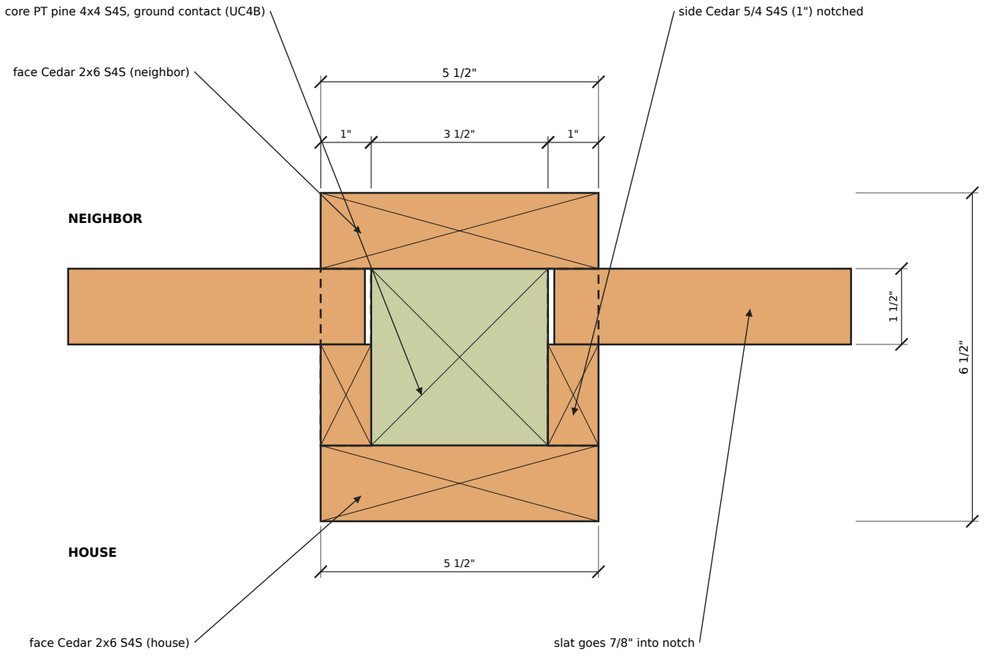
## Sections and details

Date: 2026-11-10 Rev.: 7 By:

Dimensions in feet/inches. Lumber: actual sizes (2x6 = 1 1/2" x 5 1/2").

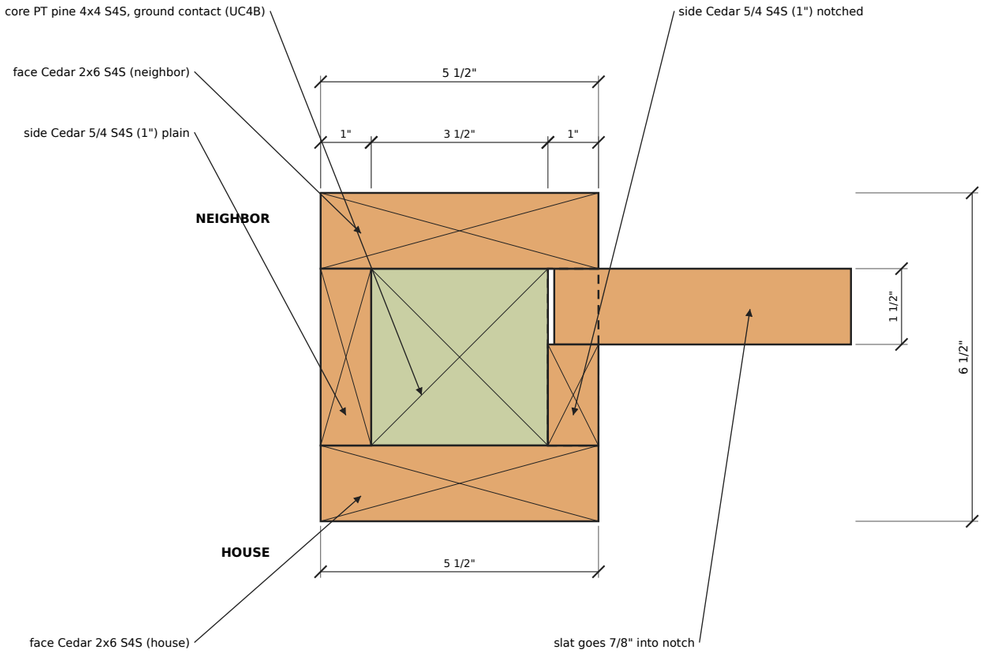
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**A-2**




## DETAIL 2a — INTERMEDIATE POST, SECTION B-B

SCALE 1:3




## DETAIL 2b — END POST (1 and 10)

SCALE 1:3

| # | INSTALLATION SEQUENCE |
|---|---|
| 1 | HOLES: lay out 10 post centers at 2'-4 13/16" o.c.; dig Ø12" x 2'-6" with rented machine auger; 6" gravel. |
| 2 | POSTS: set PT 4x4 cores 2'-0" deep, plumb and brace, pour concrete with sloped collar. Cure 48 h. |
| 3 | LEVEL TOPS: cut all cores to 7'-8" above grade (laser level). |
| 4 | NOTCHED SIDE BOARDS (pre-cut): fasten to core sides from the neighbor side, notches facing neighbor. |
| 5 | SLATS (pre-cut): drop into notches from the neighbor side, bottom to top; 1 screw per end. |
| 6 | FRONT CLOSURE: 2x6 face boards on both faces (closes the notches and locks the slats). |
| 7 | SIDE CLOSURE: 1" plain side board on the outer side of end posts 1 and 10. |
| 8 | TOP BOARDS (pre-notched): 2x8 each side of the posts, 1/2" thru-bolt per post; splices over posts 4, 7. |
| 9 | END CAPS (pre-cut): close the ends of the top boards for a solid-beam look. |
| 10 | PERGOLA RAFTERS (pre-cut): drop into top-board notches, 1 per post; hidden trim-head screws. |

**Kitchen Privacy Screen — 10 posts, cedar with pergola top (Cornelius, NC)**

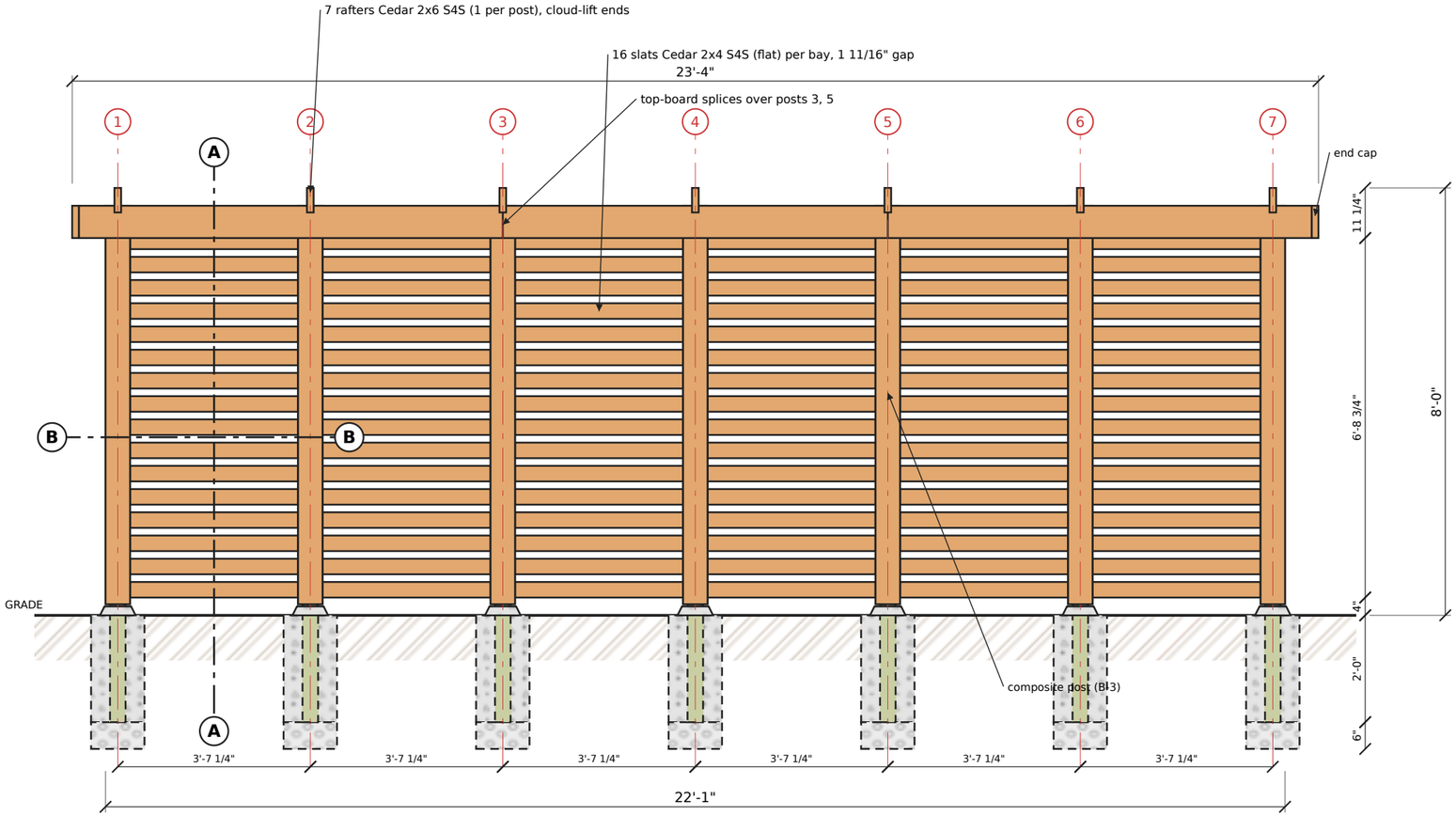
## Composite post and installation

Date: 2026-11-10 Rev.: 7 By:

Dimensions in feet/inches. Lumber: actual sizes (2x6 = 1 1/2" x 5 1/2").

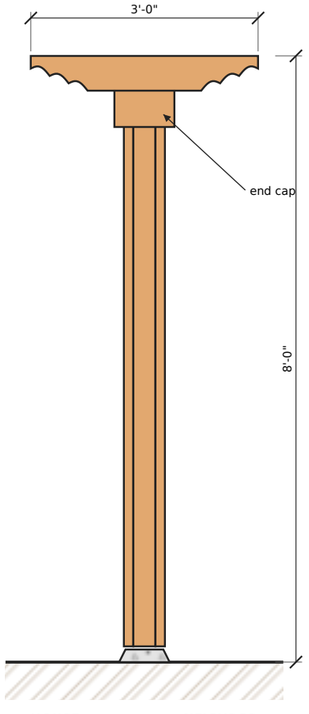
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**A-3**



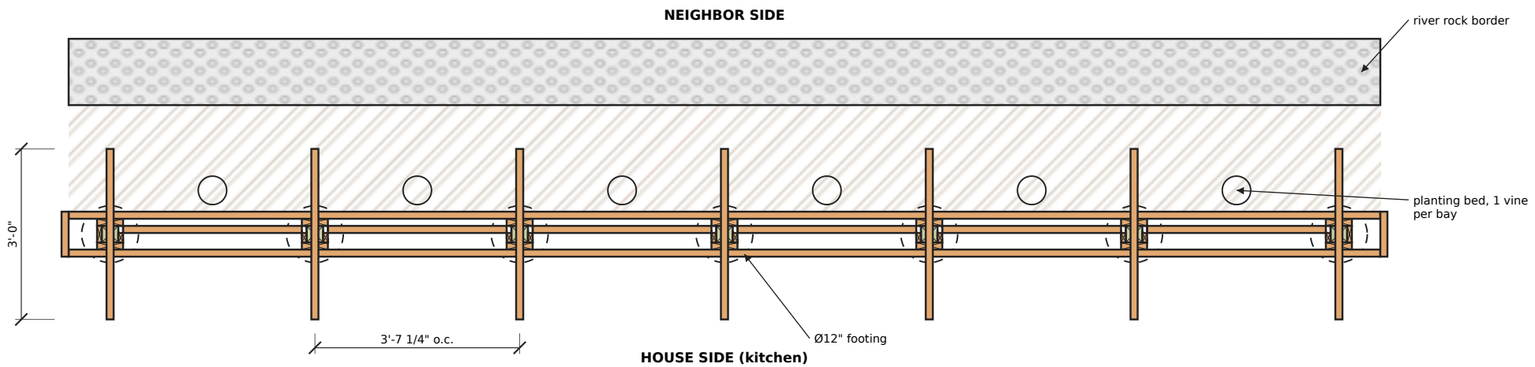

## ELEVATION — NEIGHBOR SIDE

SCALE 1:32 (3/8" = 1'-0")




## SIDE ELEVATION (end view)

SCALE 1:24 (1/2" = 1'-0")




## PLAN — TOP VIEW

SCALE 1:32 (3/8" = 1'-0")

**NOTES**

1. Overall length 22'-1"; 7 posts at 3'-7 1/4" o.c.; clear bay 3'-1 3/4".
2. Bay/post = 6.86 ≈ φ^4 (Fibonacci 89/13); span 3'-1 3/4" > 36" limit (< 48" max).
3. Post cores embedded 2'-0" in concrete over 6" gravel; holes by machine auger.
4. Call NC811 before digging. Verify property line, setbacks and HOA.
5. Fasteners: coated trim-head deck screws (ACQ-rated), countersunk.
6. 16 slats per bay, 1 11/16" gap set by the notched side boards.
7. All posts 5 1/2" wide, faces flush. Parts marked YES are pre-cut in the shop.
8. Cedar = Western Red Cedar, kiln-dried, S4S (planed 4 sides).
9. PT = Southern Yellow Pine #2, ground contact.

**Kitchen Privacy Screen — 7 posts, cedar with pergola top (Cornelius, NC)**

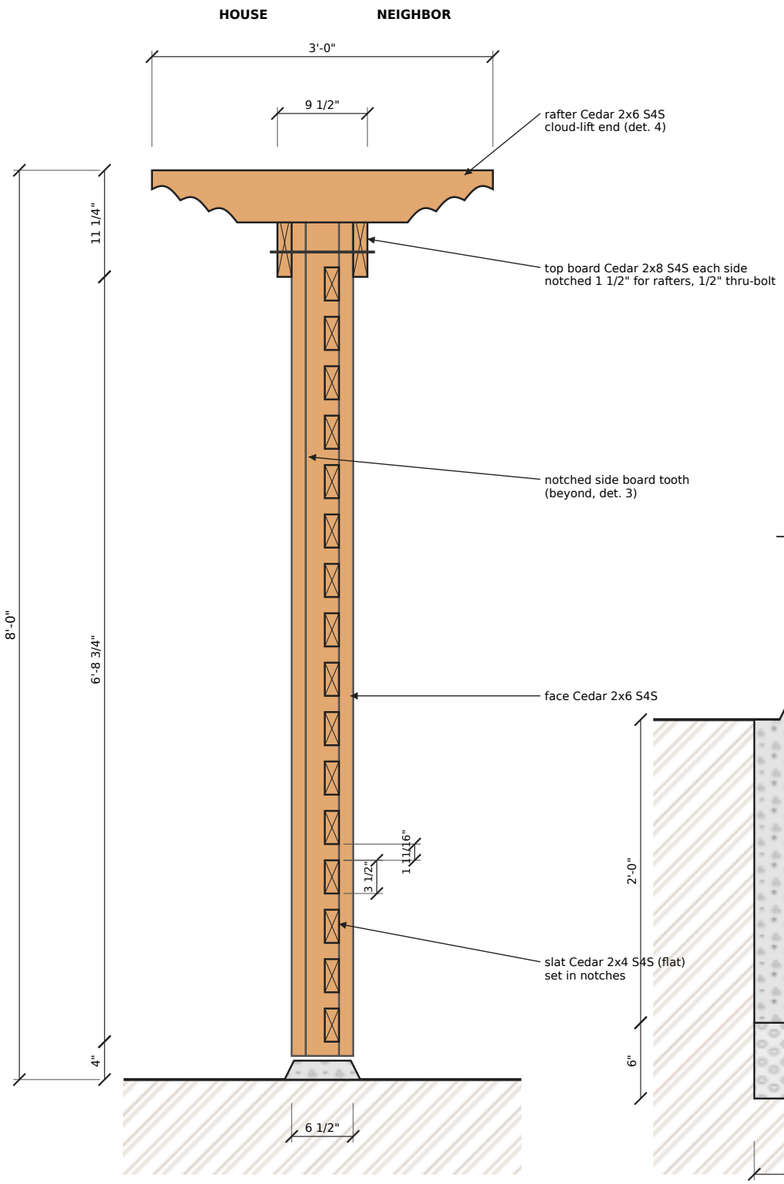
### Elevation and plan

Date: 2026-11-10 Rev.: 7 By:

Dimensions in feet/inches. Lumber: actual sizes (2x6 = 1 1/2" x 5 1/2").

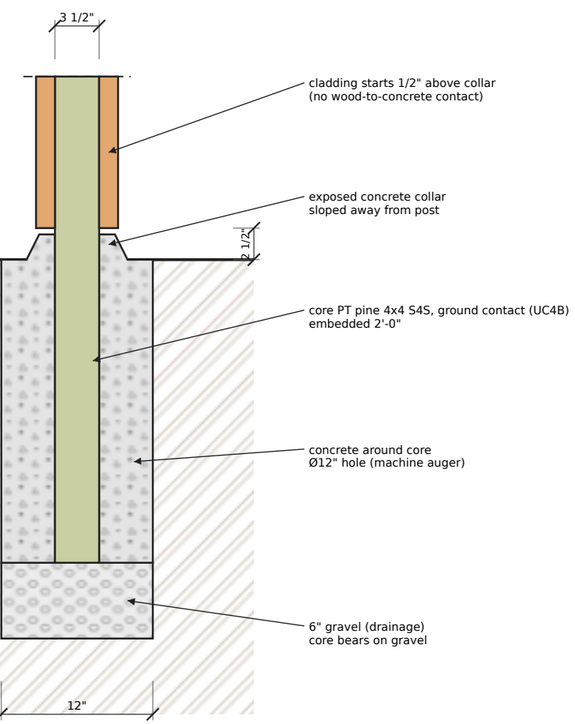
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**B-1**



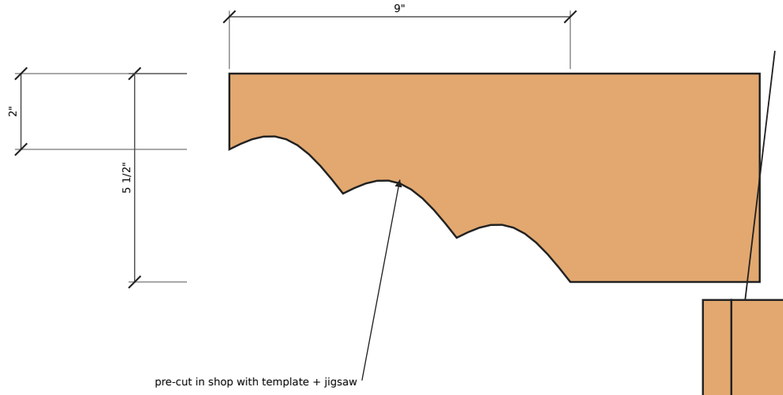

## SECTION A-A (mid-bay, looking at post 1)

SCALE 1:16 (3/4" = 1'-0")



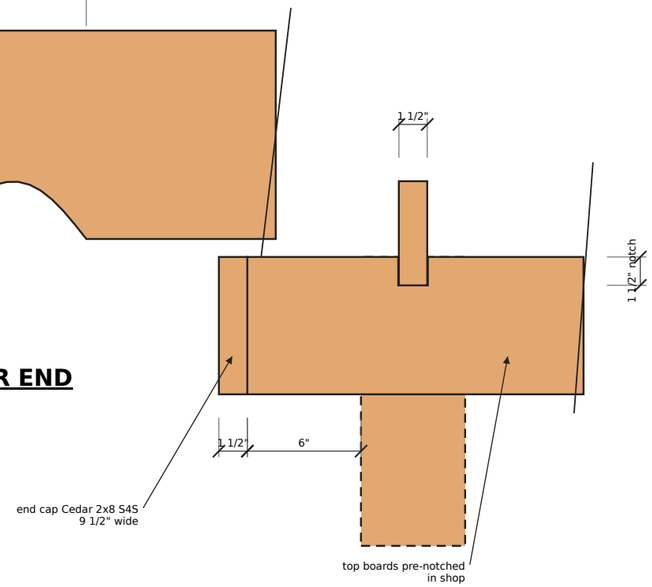

## DETAIL 1 — POST FOOTING

SCALE 1:12 (1" = 1'-0")



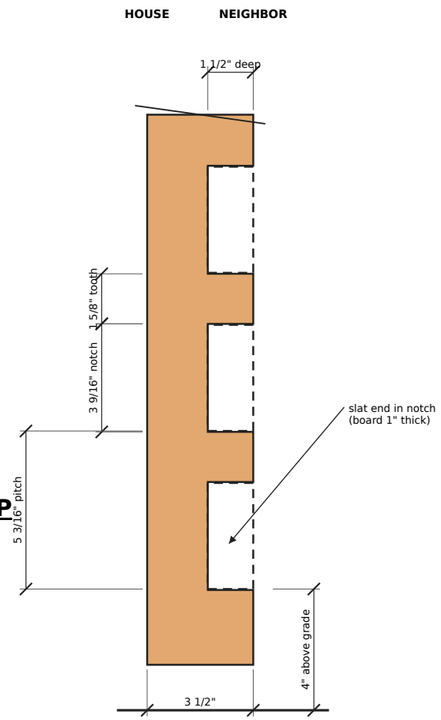

## DETAIL 4 — RAFTER END

SCALE 1:4 (3" = 1'-0")




## DETAIL 5 — TOP BOARD END: NOTCH + END CAP

SCALE 1:8 (1 1/2" = 1'-0")




## DETAIL 3 — NOTCHED SIDE BOARD (from bay)

SCALE 1:5

**PARTS LIST**

| Mark | Part | Material | Qty | Length | Shop pre-cut / notes |
|---|---|---|---|---|---|
| N | Post core | PT pine 4x4 S4S, ground contact (UC4B) | 7 | 9'-8" | embed 2'-0"; trim tops on site |
| S | Notched side board | Cedar 5/4 S4S (1") notched | 12 | 7'-5 1/2" | YES: rip to 3 1/2", 16 notches |
| R | Slat | Cedar 2x4 S4S (flat) | 96 | 3'-3 1/2" | YES: cut to length (stop block) |
| F | Post face | Cedar 2x6 S4S | 14 | 7'-5 1/2" | YES: cut to length |
| L | Plain side (end posts) | Cedar 5/4 S4S (1") plain | 2 | 7'-5 1/2" | YES: rip to 3 1/2" |
| T | Top board (3 per side) | Cedar 2x8 S4S | 6 | 7'-11 1/4", 7'-2 1/2" | YES: notches for rafters |
| E | Top board end cap | Cedar 2x8 S4S | 2 | 9 1/2" | YES |
| K | Pergola rafter | Cedar 2x6 S4S | 7 | 3'-0" | YES: cloud-lift ends |

**Kitchen Privacy Screen — 7 posts, cedar with pergola top (Cornelius, NC)**

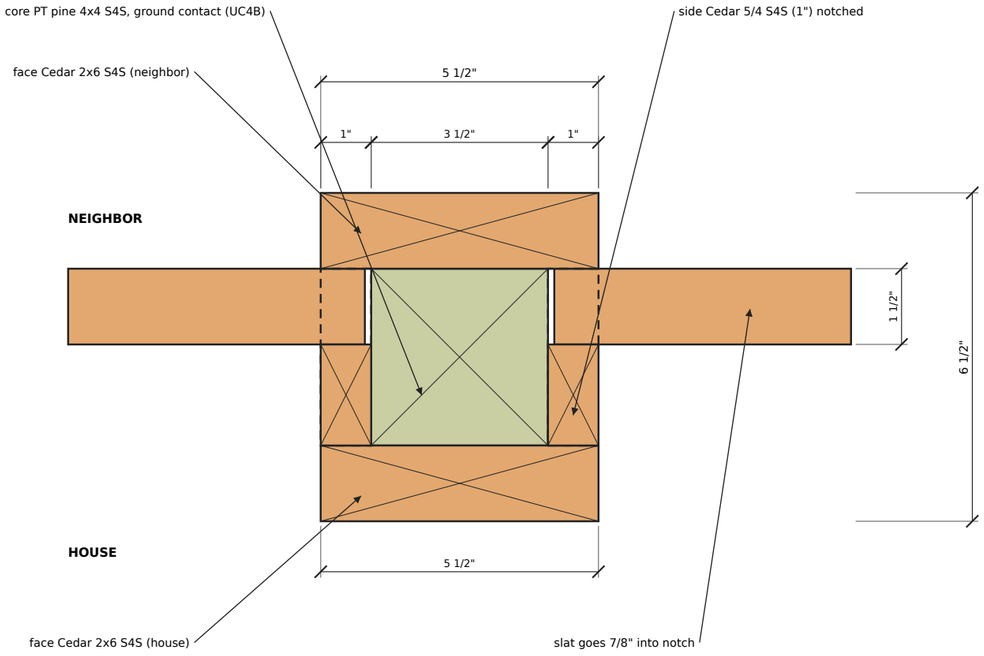
## Sections and details

Date: 2026-11-10 Rev.: 7 By:

Dimensions in feet/inches. Lumber: actual sizes (2x6 = 1 1/2" x 5 1/2").

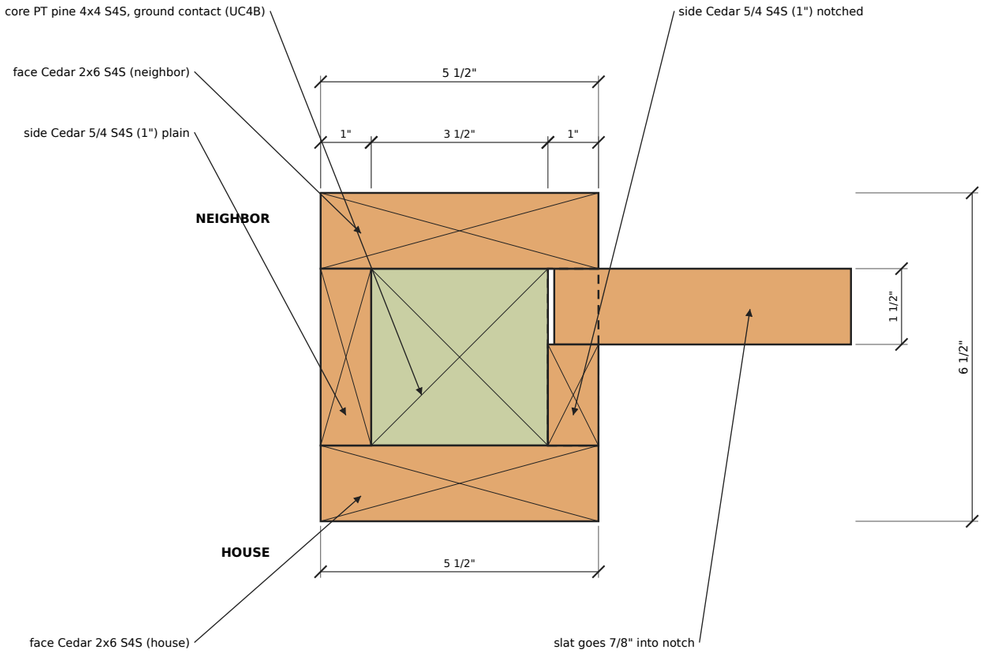
Verify all dimensions in the field before cutting. Not a structural design.

**B-2**




## DETAIL 2a — INTERMEDIATE POST, SECTION B-B

SCALE 1:3




## DETAIL 2b — END POST (1 and 7)

SCALE 1:3

| # | INSTALLATION SEQUENCE |
|---|---|
| 1 | HOLES: lay out 7 post centers at 3'-7 1/4" o.c.; dig Ø12" x 2'-6" with rented machine auger; 6" gravel. |
| 2 | POSTS: set PT 4x4 cores 2'-0" deep, plumb and brace, pour concrete with sloped collar. Cure 48 h. |
| 3 | LEVEL TOPS: cut all cores to 7'-8" above grade (laser level). |
| 4 | NOTCHED SIDE BOARDS (pre-cut): fasten to core sides from the neighbor side, notches facing neighbor. |
| 5 | SLATS (pre-cut): drop into notches from the neighbor side, bottom to top; 1 screw per end. |
| 6 | FRONT CLOSURE: 2x6 face boards on both faces (closes the notches and locks the slats). |
| 7 | SIDE CLOSURE: 1" plain side board on the outer side of end posts 1 and 7. |
| 8 | TOP BOARDS (pre-notched): 2x8 each side of the posts, 1/2" thru-bolt per post; splices over posts 3, 5. |
| 9 | END CAPS (pre-cut): close the ends of the top boards for a solid-beam look. |
| 10 | PERGOLA RAFTERS (pre-cut): drop into top-board notches, 1 per post; hidden trim-head screws. |

**Kitchen Privacy Screen — 7 posts, cedar with pergola top (Cornelius, NC)**

## Composite post and installation

Date: 2026-11-10 Rev.: 7 By:

Dimensions in feet/inches. Lumber: actual sizes (2x6 = 1 1/2" x 5 1/2").

Verify all dimensions in the field before cutting. Not a structural design.

**B-3**